Protecting Salem's Ecosystem:

Predictive Modeling of Tree Canopies and Stream Temperatures

Siera Edwards, Serenna Walter, and Brooke Proctor

Research Questions

1. Can we use historic stream temperature data to predict stream temperature 3 years in the future?

2. What are the existing patterns and anomalies in the tree canopy coverage along Salem's streams?
3. What are the existing patterns and anomalies in the historic weather data for Salem, Oregon?
    a. This will utilize only the weather_temperature_salem data to construct a model to predict weather and to predict anomalies that may point toward hazardous conditions for trees in riparian zones.
4. Can these additional extrapolated metrics improve the performance of our predictive model for stream temperature?

Data Inventory

Source	**Volume**	**Coverage**	**Access**
Ds_daily.csv Source: Mill Creek Research Project by CastawayWU Available by request.	Rows: 7543 Columns: 5 Bytes: 426 KB	Date: 2014-12-31 to 2025-6-4 Streams: Mill Creek Data collection sites: MIC12 (closer to the	Request data from Dr. Gore.

		industrial center of Salem), MIC3 (close to campus)	
Public_Storm_Creeks_and_Rivers.shp Source: City of Salem Oregon GIS	Shapefile, rows: 2,672, columns: 24, bytes: 619 KB	Contains shapes for all streams in Salem. This will allow us to determine which trees are nearest to bodies of water. Last updated: May 30, 2026	Public and Downloaded. The file is in data/raw License: The data in SalemMaps Online has been prepared for use by GIS professionals and as a reference map for online users. For official City of Salem information and maps, please go to https://www.cityofsalem.net
Weather Data: 4346262.csv Source: NOAA	Rows: 48603 Columns: 18 Bytes: 15.1 MB	Has the daily temperature max and min, as well as additional data information. This spans from 1900-01-01 to 2026-06-30.	Publicly available, but must be requested from the NOAA website. Requests take maybe five minutes from opening the website to receiving the data. Currently downloaded.

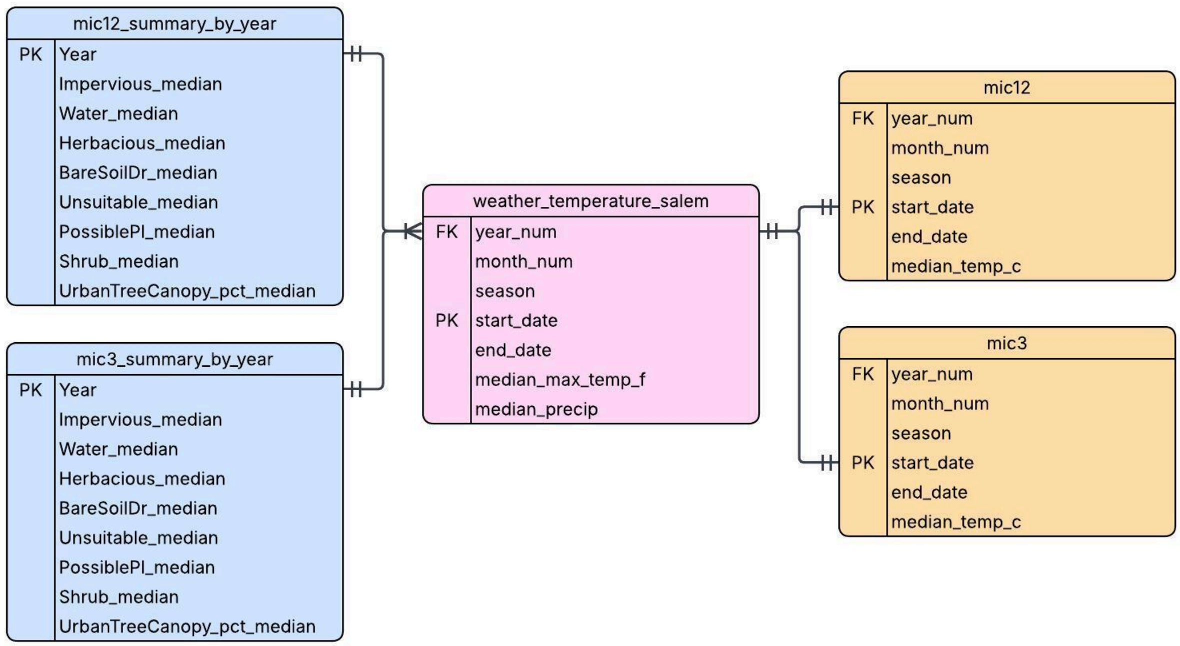
Tree Canopy: raw_urban_tree_canopy.shp Source: [Tree Plotter Canopy]	Rows: 3,649 Columns (attribute table): 18 Bytes: 3,184 KB	Has polygons for different types of tree canopy areas across Salem.	Publicly available map and user interface, but must have a login from the City of Salem to download the data. Can request data from the City of Salem.

We do not intend to refresh this data or continuously scrape throughout the process. We will utilize stagnant files.

All collected data is being stored in the Railway database and is accessible to each group member through an SQL client. The Railway database is notably not a normalized relational database because several of the data files are not being joined. The use of the database is mostly to avoid downloading files on each team member's computer.

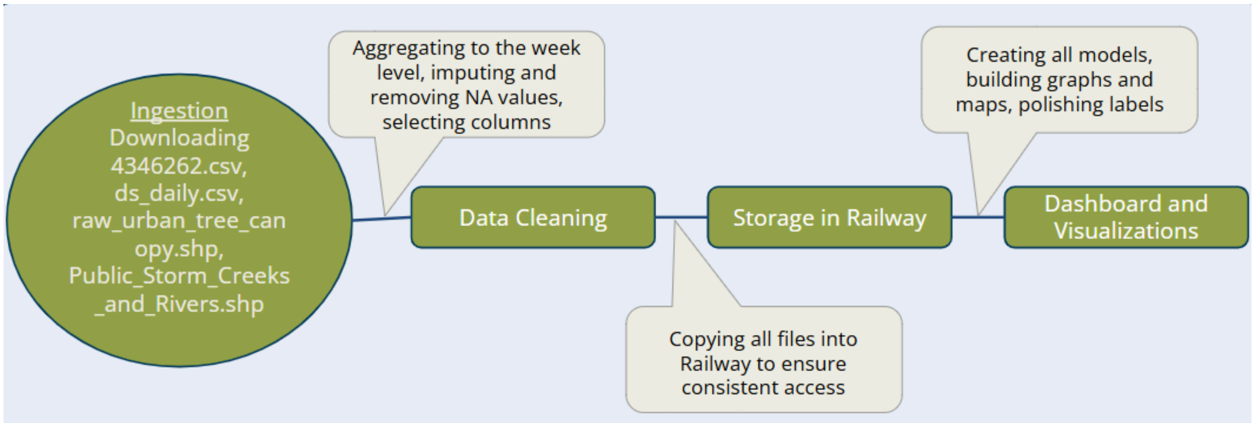
Schema and Organization

Entity Relationship Diagram



The one-to-one relationships in weather_temperature_salem to mic12 and mic3 indicated that these could exist in one large table, but we have opted to keep them separate for division of labor purposes. Additionally, the color groupings of this indicate which tables came from the same initial CSV files.

Stages and Transformations



Ingestion Status & Data Quality and Proofing

Note: Our data cleaning process and data proofing inspection for each dataset are in one RMarkdown file, Capstone_Data_Cleanup.RMD, which includes sections for each dataset.

Stream Temperature and Dissolved Oxygen

Original Dataset: ds_daily.csv

Cleaned Dataset: mic12.csv, mic3.csv

1. Ingestion
    a. Pulling the Data:

       This secondary stream temperature dataset was pulled by Dr. Gore at Willamette University and shared with our group. This data was collected for the Mill Creek Research Project by CastawayWU. It is recommended to update this dataset when new data is available by requesting it from the leaders of CastawayWU or contacting Dr.Gore.

b. Cleaning:

After reading in this dataset into our RMarkdown for cleaning our data, it was discovered that the only cleaning needed were null values. Since we are planning to aggregate the data to the month level, removing these rows entirely may have negative downstream effects. Instead, we decided to impute the NA values with the median value of the avg_do column. The stream temperature column did not have any null values but the stream dissolved oxygen contained 102 null values so 102 rows were imputed. Then the data was separated into two datasets, one for each data collection site. Columns with values for the year number, month number, and the season of the data collection date. The date range was then filtered down to the start of 2016 to the end of 2024. Finally, the datasets were both aggregated to the month level using the xts package with the median stream temperature being retained. Finally, the datasets were both written to CSV files to be used in our analysis.

c. Adding Data to the Database:

This data was moved into the Railway Postgres and used in our Python file to build our prediction models. Some graphs were made in R to depict trends in this data and they were saved as a PNG file and moved to the data/processed/ file in our project GitHub.

d. Evidence of Success:

The graphs were created and do, in fact, show the relationship that scientists have discovered between stream temperature and dissolved oxygen along with other trends in the data.

2. Data Quality and Proofing

    a. Missingness:

    This dataset did not have any missing values in the date_only, Location, and avg_temp columns. However, it did contain 102 NA values in the avg_do column and 830 NA values in the avg_turbidity column.

    b. Duplicates:

    There were no duplicate rows. Each day and site had exactly one record in the dataset.

    c. Outliers:

    There were no significant outliers in this dataset. This was confirmed by graphing the relationship between average daily temperature and average daily dissolved oxygen. None of the values appeared to be significantly differing from the strong relationship shown. Additionally, the minimum, maximum, and median values were examined using summary(df) and none were inconsistent with expectations.

    d. Label Issues:

    There were no significant labeling issues with this data so the column names were left as they were. The only label adjustments happened in graph labels for additional clarity.

Streams in Salem

Original Dataset: Public_Storm_Creeks_and_Rivers.shp

Cleaned Final Dataset: creek_locations.shp and creek_locations.csv

I. Ingestion

    A. Pulling the Data:

This dataset was pulled directly from the Public Storm Creeks and Rivers map on the City of Salem Oregon GIS website. This dataset is historical, so the data is pulled manually. It is recommended to update this dataset when new data is available.

B. Cleaning:

To prepare this shapefile for further use, we simply had to remove the columns that did not contain information relevant to our research. There were many columns in this dataset that were primarily null values, so these were removed using the select statement to create a smaller and more focused shapefile. The changes were saved to the creek_locations.shp file. The same columns were removed for the CSV, leaving the final cleaned CSV for this dataset, creek_locations.csv. This file is now ready to be joined with other files to perform analysis.

C. Adding Data to the Database:

The final cleaned CSV, creek_locations.csv, was ingested into PostgreSQL and into our Railway project “Emerald Ash Borer Prevention”. The other files and the shapefile are added to the repository.

D. Evidence of Success:

When comparing the row counts from the original csv to the table in Railway, we can see that the row counts are consistent with 2672 rows.

II. Data Quality and Proofing

A. Missingness

This dataset has a lot of missing values in the columns we selected, but we determined that this was not a problem because these columns are not the ones we are depending on for modeling. The columns that include a lot of missing values are: name, maintained_by, and pw_grid_num.

B. Duplicates

There are no duplication errors for this dataset.

C. Outliers

Outliers don't really apply to this dataset since this is a shapefile map of the streams in Salem.

D. Label Issues

We renamed the columns in the CSV to be all lowercase and the separate words separated by underscores. For example: "PW Grid No" was changed to pw_grid_num, UnitID was changed to unit_id, etc.

Weather

Original Dataset: USW00024232.csv

Cleaned Final Dataset: weather_temperature_salem.csv

II. Ingestion

E. Pulling the Data:

This data was downloaded from a web pull from [NOAA](). This process had various steps to filter and select the data. The web pull settings we used to get the original dataset for our project are as follows:

1. Type of dataset: Daily Summaries

2. Date Range: 1900-01-01 to 2026-06-30
3. Stations (Search): McNary Field

   Click to download the csv

F. Cleaning:

   The raw file, USW00024232.csv, contained many additional columns that were not needed for our analysis. The main cleaning task was selecting relevant columns to use in analysis with a select() statement. The columns chosen for analysis were STATION, NAME, LATITUDE, LONGITUDE, DATE, AWND, PRCP, SNOW, SNWD, TMAX, TMIN, and TAVG. After that, NAs in TMAX, TMIN, PRCP, and DATE were dropped. This gives us adequate information about weather, temperature, precipitation, and the collection site, and NAs are no longer an issue for model creation. Additionally, this information was aggregated using a groupby/summarize() to give us the daily, monthly, and seasonal levels of observation.

   a. Note: For modeling, the date range only includes our research question years 2020, 2022, and 2024. This is easy to filter, so we left our cleaned dataset with the wide year range because we may want to include visualizations that show more historical context.

G. Adding Data to the Database:

   The final cleaned CSV, weather_temperature_salem.csv, was ingested into PostgreSQL and into our Railway project “Emerald Ash Borer Prevention”.

H. Evidence of Success:

This is a very large dataset. When comparing the row counts from the original CSV to the table in Railway, we can see that the row counts are consistent with 45723 rows.

III. Data Quality and Proofing

A. Missingness

This dataset has a lot of missing values, but it is also very large. We removed rows with missingness in our variables of interest–TMIN, TMAX, DATE, and PRCP. With the missing variables gone, the variables that have remaining missingness are as follows:

1. Temperature Average: 43,047 (almost all observations)
2. Average Wind Speed: 30,215
3. Snow Depth: 20,647
4. Snow: 15,848

Something to note about this dataset is that the null values don't mean that the values here should be zero. Zero is entered when there is a record for the different weather recording variables, not null. We are not entirely sure why these null values occur, but a majority seem to come from older dates than newer ones.

B. Duplicates

No duplicated rows were found in this dataset.

C. Outliers

When examining the different weather variables, we can see that there are outliers seen many times throughout history in precipitation, snow, snow depth, wind speed, and temperature. We decided not to remove these outliers yet in this process because we want to see if they will have an impact, if any, on our modeling. We anticipate very little impact since we have such a large time range.

D. Label Issues

All variables in the summarized dataframe were relabeled to be more readable: year_num, month_num, season, start_date, end_date, median_max_temp_f, median_precip are the appropriate labels for the numeric year, numeric month, start date of the month the data was observed in, end date of the month the data was observed in, median value of the maximum temperature in Fahrenheit, and the median precipitation amount in inches, accordingly.

Tree Canopy

Original Dataset: raw_urban_tree_canopy.shp and creek_locations.shp

Cleaned Final Dataset: mic3_summary_by_year.csv, mic12_summary_by_year.csv, tree_canopy_mic3.csv, and tree_canopy_mic12.csv

III. Ingestion

E. Pulling the Data:

This data was downloaded from the Tree Canopy website, using the login from Meredith, who works for the City of Salem.

F. Cleaning:

The process for generating the final tree canopy dataset uses two raw files: the cleaned creek locations shapefile (creek_locations.shp) and the raw tree canopy shapefile downloaded from the Tree Canopy website (raw_urban_tree_canopy.shp). Both of these shapefiles were put into ArcGIS Pro.

First, the Buffer tool in ArcGIS Pro was used to create a 100ft buffer on the streams to represent the riparian zone. This 100ft buffer represents the geographic areas we are going to focus on for tree canopy along streamsides. Next, this buffer layer was used in the Clip tool to clip the tree canopy shapefile so that the data for tree canopy only includes the streamside riparian zones. Next, the Selection tool was used to select only the stream represented by the MIC3 and MIC12 stream temperature measurement sites. The stream segments corresponding to each of these measurement sites are of similar length. From this layer, the attribute table was exported. Separate attribute tables for MIC3 and MIC12 were exported as well so that the two creek segments can be modeled separately. All of these files are attached in the repository for easy reference as each step was conducted in ArcGIS Pro. After the GIS processing was complete, the MIC3 and MIC12 tables were wrangled in R to form the final datasets for both creek segments. The tables were converted into long format, creating the variable UrbanTreeCanopy_pct as the percentage of urban tree canopy as observed that year. Unnecessary columns were removed, keeping 9 columns: Impervious, Water, Herbaceous, BareSoilDr, Unsuitable, PossiblePl, Shrub, Year, and UrbanTreeCanopy_pct. This was then used to create median aggregates for each year for all the variables for MIC3 and MIC12 creek locations, creating two final modeling datasets: mic3_summary_by_year and mic12_summary_by_year. For other visualization purposes, the non-aggregated versions of the datasets were also saved: tree_canopy_mic3.csv and tree_canopy_mic12.csv.

G. Adding Data to the Database:

Both the modeling aggregated datasets and the non-aggregated versions are added to the GitHub repository. For other visualization purposes, the non-aggregated datasets, tree_canopy_mic3.csv and tree_canopy_mic12.csv, were ingested into PostgreSQL and into our Railway project "Emerald Ash Borer Prevention".

H. Evidence of Success:

When comparing the row counts from the original CSVs to the table in Railway, we can see that the row counts are consistent with 3 rows for each mic3_summary_by_year and mic12_summary_by_year (3 years, 3 rows).

IV. Data Quality and Proofing

A. Missingness

There are no missing values for this dataset.

B. Duplicates

There is no duplication for this dataset.

C. Outliers

There are no obvious outliers in this dataset.

D. Label Issues

There were no labeling issues for the columns in this dataset. The labels were straightforward. We did not have a lot of documentation on these variables in this dataset, however.

Reproducibility

Python version 3.11.1 was used for this project, as well as R version 4.4.1 and 4.5.2.

We have combined all of our data cleaning and modification into an RMarkdown titled "Capstone_Stream_Data_Cleanup.Rmd". This file takes in the files

- ds_daily.csv
- Public_Storm_Creeks_and_Rivers.shp
- 4346262.csv
- creekside_tree_canopy_MIC3.csv
- creekside_tree_canopy_MIC12.csv

After cleanup and modification, it creates the files:

- mic3.csv
- mic12.csv
- creek_locations.shp
- Weather_temperature_salem.csv
- mic3_summary_by_year.csv
- mic12_summary_by_year.csv
- tree_canopy_mic3.csv
- tree_canopy_mic12.csv

After this data cleaning and collection, the final files are moved into a PostgreSQL database hosted by Railway. In this PostgreSQL database, the files will not be normalized. We are not necessarily trying to build a relational database.

We are using Railway primarily as a tool to share data amongst the three of us without downloading each file. Because the data has been cleaned using an R script, we are only warehousing the cleaned, aggregated data files in Railway. This will also support the dashboard that is created as a part of our platforming and visualization stage.

Ethics and Access Controls

Currently, our data contains no PII, so we are not concerned about accidentally sharing people's personal information. All of our data is publicly available, so we are not concerned about consent or redaction. Our main ethical concern with this project is creating predictions with the limited years of data that we have access to. Since we only have tree canopy data for 2020, 2022, and 2024, we are only creating models for these years. This risks missing overall trends that are seen across years. This also limits our ability to make predictions into the future, rather we must focus on emphasising the scope of stream temperature change in Salem between years. We will take extra precautions by ensuring that our conclusions are phrased properly and do not overstate our findings to prevent inaccurate and harmful decisions made with our findings. We will also share these concerns in our final presentation to make the audience aware of these limitations to our predictions' usability.

Freeze Statement

No further data ingestion is planned. We feel confident that this data will be able to answer our research questions.

Our project has changed slightly since our most recent project proposal. After our expert consultation with Professor Cheng, we came to the conclusion that we had to alter our research questions slightly. Instead of focusing on predicting stream temperatures into the future, our lack of historical data for tree canopy data pushed us to focus on modeling the years that we have data for and find trends within that. This way, we can identify historical trends and any changes over the year and still report reliable findings. This slightly affected which datasets we decided to pursue in the analysis but we didn't add any new ones.